

Fractional Prepayment Games in Financial Networks

A reconstruction and extension of the 2020 LMFI notes

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Abstract

This note develops the fractional-prepayment direction suggested by a set of 2020 working notes on clearing in interbank debt networks. The underlying model is the Rogers–Veraart extension of Eisenberg–Noe, with losses on external assets and interbank receipts in default. Before clearing, each bank may use liquid external assets to repay any fraction of an existing liability. The resulting game is a continuous counterpart of the all-or-nothing prepayment game introduced in the later literature.

The first contribution is structural. On every regular region with a fixed default set, clearing payments are rational, and hence differentiable, functions of selective prepayments. They are affine when prepayments preserve each debtor’s liability proportions. An explicit resolvent formula gives the Jacobian. The second contribution is local mechanism design. For one insolvent debtor, the note derives the exact cone of infinitesimal repayment directions that weakly benefit every creditor; pro-rata repayment is the unique direction that preserves all residual liability shares. The third contribution is strategic. A four-node example has a unique best response that prepays a strict fraction of one liability, showing that fractional strategies cannot in general be reduced to all-or-nothing choices. Earlier results from the notes on fusion and round-trip loans are retained as background.

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1 Purpose, contribution, and related literature

The original notes move between a clearing model, comparative statics, and a game in which banks alter their balance sheets before clearing. Several calculations are exploratory, and some statements labelled as theorems are followed by requests to verify them. The aim here is to make the assumptions visible and to pursue the strongest undeveloped direction: continuous prepayment before clearing.

The main conclusions are as follows.

- (i) The greatest clearing fixed point is the natural selection when bank value is increasing in receipts from the network.
- (ii) Fusion has a transparent local accounting rule. The deficit of the fused node is the sum of the two original deficits and the fusion cost, minus the two unpaid internal claims that disappear in the fusion.
- (iii) In the restricted three-node model, balance-sheet-consistent round-trip lending cannot increase the solvent accomplice's payoff. The conclusion in the thesis is valid, but the asymmetric payoff formula needs a cash-flow correction.
- (iv) Selective early repayment creates aggregate value when liquidation of external assets is costly. The selected creditor weakly gains, but other creditors may lose. The exact local Pareto cone describes how much must be repaid to each creditor to eliminate this conflict.
- (v) A continuous prepayment game has compact polyhedral strategy sets. Within a regular fixed-default region its payoffs are differentiable rational functions. A resolvent formula reduces every local marginal calculation to one linear solve.

- (vi) Fractional actions are essential. There are networks in which a player's unique best response repays neither zero nor the whole liability. The optimum is the least payment that moves another bank across a solvency boundary.

Statements about derivatives are local to a region in which the same banks remain solvent or insolvent. Crossing a default boundary changes the formula for the clearing map and must be handled separately.

1.1 Relation to prior work

Eisenberg and Noe established the fixed-point clearing model with proportionality, limited liability, and priority of debt [1]. Rogers and Veraart introduced default costs and studied mergers and rescue consortia [2]. The fusion material below therefore sharpens an existing operation rather than proposing a new one. Netting and portfolio compression had also shown before 2020 that removing gross or circular liabilities can have adverse distributional and systemic effects [3, 4].

Strategic departures from proportional clearing were studied by Bertschinger, Hoefer, and Schmand [5]. Papp and Wattenhofer showed that debt forgiveness and transfers of external assets can improve a bank's payoff through network feedback [6]. These operations are close to prepayment, but they do not model the same balance-sheet update at an earlier date. Work on multiple maturities supplies a dynamic clearing background [7].

Sensitivity formulas for Eisenberg–Noe payments predate the present note. Liu and Staum differentiate with respect to external assets [8]; Feinstein et al. differentiate with respect to bilateral liabilities [9]. The derivative formula below applies the same general idea to the simultaneous change in assets, nominal debts, and relative liabilities caused by prepayment.

Zhou et al. later introduced a strategic prepayment game based on all-or-nothing repayment of whole liabilities [10]. Their conclusion lists fractional prepayments and coalitional incentives as open directions. The 2020 notes appear to anticipate this topic, but the present paper makes no claim of publication priority. Its narrower objective is to develop the continuous model and results that are absent from the discrete formulation.

2 Clearing with default costs

2.1 Debt networks

Let the set of banks be $\mathcal{N} = \{1, \dots, n\}$. A debt network consists of external assets $e \in \mathbb{R}_+^n$ and a liability matrix $L \in \mathbb{R}_+^{n \times n}$ with $L_{ii} = 0$. The entry L_{ij} is the nominal amount owed by bank i to bank j . Define total nominal liabilities and relative liabilities by

$$\bar{p}_i = \sum_{j \in \mathcal{N}} L_{ij}, \quad \pi_{ij} = \begin{cases} L_{ij}/\bar{p}_i, & \bar{p}_i > 0, \\ 0, & \bar{p}_i = 0. \end{cases} \quad (1)$$

Thus $\pi_{ij}p_i$ is the payment from i to j when i makes the total payment p_i .

Let $\alpha, \beta \in [0, 1]$. The parameter α is the recovery fraction on external assets in default, and β is the recovery fraction on interbank receipts. The Rogers–Veraart clearing map $\Phi : [0, \bar{p}] \rightarrow [0, \bar{p}]$ is

$$\Phi_i(p) = \begin{cases} \bar{p}_i, & e_i + \sum_{j \in \mathcal{N}} \pi_{ji}p_j \geq \bar{p}_i, \\ \alpha e_i + \beta \sum_{j \in \mathcal{N}} \pi_{ji}p_j, & e_i + \sum_{j \in \mathcal{N}} \pi_{ji}p_j < \bar{p}_i. \end{cases} \quad (2)$$

A clearing vector is a fixed point of Φ .

The map is monotone in the product order. Tarski's fixed-point theorem gives a least and a greatest fixed point. We denote the greatest one by $p^\uparrow$. It can also be obtained by the decreasing iteration or default-set algorithm described by Rogers and Veraart [2].

2.2 Why select the greatest fixed point?

For a payment vector p , define the post-clearing equity of bank i by

$$V_i(p) = \left[e_i + \sum_{j \in \mathcal{N}} \pi_{ji} p_j - \bar{p}_i \right]^+. \quad (3)$$

This definition treats debt payments as senior to equity.

Proposition 2.1 (Pareto dominance of the greatest fixed point). *For every clearing vector p , one has*

$$V_i(p^\uparrow) \geq V_i(p) \quad \text{for every } i \in \mathcal{N}. \quad (4)$$

Proof. By definition of the greatest fixed point, $p^\uparrow \geq p$ coordinatewise. Every coefficient π_{ji} is nonnegative, so the expression inside the positive part in (3) is weakly increasing in p . The positive part is also increasing. $\square$

The selection matters when default costs destroy uniqueness. For example, take two banks with

$$L_{12} = L_{21} = 2.2, \quad e_1 = e_2 = 1, \quad \alpha = \beta = 0.5. \quad (5)$$

Both $(1, 1)$ and $(2.2, 2.2)$ are clearing vectors. The second is the greatest fixed point. This example also shows why an iteration from zero cannot simply be interpreted as selecting the economically preferred outcome.

3 Fusion and the cancellation of internal claims

3.1 Definition of fusion

Fix two distinct banks $a, b \in \mathcal{N}$. Their fusion is a new node $[ab]$. Mutual liabilities L_{ab} and L_{ba} disappear, while liabilities to and from all other banks are added. If $\kappa \geq 0$ is the total fusion cost, then

$$e_{[ab]} = e_a + e_b - \kappa, \quad (6)$$

$$\bar{p}_{[ab]} = \bar{p}_a + \bar{p}_b - L_{ab} - L_{ba}, \quad (7)$$

$$L_{[ab]j} = L_{aj} + L_{bj}, \quad j \notin \{a, b\}, \quad (8)$$

$$L_{j[ab]} = L_{ja} + L_{jb}, \quad j \notin \{a, b\}. \quad (9)$$

The full clearing vector may change after fusion because the fused node has a new liability proportion. It is therefore useful to begin with a local identity at fixed external payments.

3.2 A fusion deficit identity

For any candidate payment vector $p \in [0, \bar{p}]$, define the pre-haircut resource deficit of bank i by

$$\delta_i(p) = \bar{p}_i - e_i - \sum_{j \in \mathcal{N}} \pi_{ji} p_j. \quad (10)$$

Positive $\delta_i(p)$ means that i cannot pay in full at the candidate payments. Define the unpaid portions of the two internal claims by

$$u_{ab}(p) = L_{ab} - \pi_{ab} p_a, \quad u_{ba}(p) = L_{ba} - \pi_{ba} p_b. \quad (11)$$

Both quantities are nonnegative.

Theorem 3.1 (Fusion deficit identity). *Hold the payments of all banks outside $\{a, b\}$ fixed. The deficit of the fused node is*

$$\delta_{[ab]} = \delta_a(p) + \delta_b(p) + \kappa - u_{ab}(p) - u_{ba}(p). \quad (12)$$

Proof. The external inflow of the fused node is

$$\sum_{j \notin \{a, b\}} p_j (\pi_{ja} + \pi_{jb}). \quad (13)$$

Consequently,

$$\begin{aligned} \delta_{[ab]} &= \bar{p}_a + \bar{p}_b - L_{ab} - L_{ba} - (e_a + e_b - \kappa) \\ &\quad - \sum_{j \notin \{a, b\}} p_j (\pi_{ja} + \pi_{jb}). \end{aligned}$$

On the other hand,

$$\begin{aligned} \delta_a(p) + \delta_b(p) &= \bar{p}_a + \bar{p}_b - e_a - e_b \\ &\quad - \sum_{j \notin \{a, b\}} p_j (\pi_{ja} + \pi_{jb}) - \pi_{ab} p_a - \pi_{ba} p_b. \end{aligned}$$

Subtracting these expressions gives

$$\begin{aligned} \delta_{[ab]} - \delta_a(p) - \delta_b(p) &= \kappa - L_{ab} - L_{ba} + \pi_{ab} p_a + \pi_{ba} p_b \\ &= \kappa - u_{ab}(p) - u_{ba}(p). \end{aligned}$$

□

The identity makes the economics explicit. Fusion adds its direct cost, but it also deletes internal claims that were not going to be paid. Those unpaid claims are the only local source of a rescue benefit.

Corollary 3.2 (Local rescue criterion). *At fixed external payments, the fused node can pay in full if and only if*

$$u_{ab}(p) + u_{ba}(p) \geq \delta_a(p) + \delta_b(p) + \kappa. \quad (14)$$

Corollary 3.3 (Common-stage insolvency). *Suppose a and b are still nominal payers at a given stage, so that $p_a = \bar{p}_a$ and $p_b = \bar{p}_b$. If $\delta_a(p) > 0$ and $\delta_b(p) > 0$, then their fusion is insolvent at that stage for every $\kappa \geq 0$.*

Proof. Nominal payment implies $u_{ab}(p) = u_{ba}(p) = 0$. Apply Theorem 3.1. $\square$

This corollary gives a precise version of the claim that two banks becoming insolvent at the same default-set iteration cannot rescue each other by fusion. A global theorem additionally requires checking that payments of the other banks agree up to that iteration. This agreement holds under the usual same-stage construction before either fused bank reduces its payment, but it should not be assumed after their default because the fused liability shares then change.

Open question 3.4 (Global fusion comparative statics). Can one characterize the change in the greatest clearing vector after fusion using the resolvent of the default-set linear system? The identity (12) gives the direct effect. A complete answer must also capture feedback through the payments of all other defaulting banks.

4 Round-trip loans in a three-node network

4.1 The restricted model

Consider banks 1, 2, and n . Bank 1 owes $B > 0$ to bank 2 and $A \geq 0$ to bank n . Bank 2 owes $C \geq 0$ to bank 1. Bank 1 has no initial external assets. Banks 2 and n remain solvent, while bank 1 remains insolvent. The last condition is

$$C < A + B. \quad (15)$$

Suppose bank n lends $S_1 \geq 0$ to bank 1, and bank 1 lends $S_2 \in [0, S_1]$ back to bank n . Balance-sheet consistency gives bank 1 the external assets $S_1 - S_2$. Its total payment in default is

$$\alpha(S_1 - S_2) + \beta(C + S_2), \quad (16)$$

and the fraction directed to bank n is

$$q(S_1) = \frac{A + S_1}{A + B + S_1}. \quad (17)$$

The relevant net cash contribution to bank n is therefore

$$\nu(S_1, S_2) = [\alpha(S_1 - S_2) + \beta(C + S_2)]q(S_1) - S_1. \quad (18)$$

Only S_1 is subtracted. The cash S_2 borrowed by bank n is offset by bank n 's repayment of the same debt. Subtracting both S_1 and S_2 , as in the exploratory formula, counts the second flow twice.

4.2 No profitable round trip

Theorem 4.1 (Round-trip lending is unprofitable). *Under (15), for every $\alpha, \beta \in [0, 1]$ and every feasible pair $0 \leq S_2 \leq S_1$,*

$$\nu(S_1, S_2) \leq \nu(0, 0). \quad (19)$$

Proof. For fixed S_1 , the function ν is affine in S_2 , with

$$\frac{\partial \nu}{\partial S_2} = (\beta - \alpha)q(S_1). \quad (20)$$

First suppose $\beta \geq \alpha$. The maximum over $S_2 \in [0, S_1]$ is at $S_2 = S_1$. Put $S = S_1$ and

$$f(S) = \beta(S + C) \frac{A + S}{A + B + S} - S. \quad (21)$$

The derivative of the fraction before multiplication by β is positive, so $\beta \leq 1$ gives

$$f'(S) \leq \frac{B(C - A - B)}{(A + B + S)^2} < 0. \quad (22)$$

Now suppose $\beta < \alpha$. The maximum over S_2 is at $S_2 = 0$. Put

$$g(S) = (\alpha S + \beta C) \frac{A + S}{A + B + S} - S. \quad (23)$$

All coefficients multiplying α and β in $g'(S)$ are nonnegative. Replacing both parameters by 1 gives an upper bound, hence

$$g'(S) \leq \frac{B(C - A - B)}{(A + B + S)^2} < 0. \quad (24)$$

In both cases the best value is attained at $S_1 = S_2 = 0$. $\square$

Theorem 4.1 includes the symmetric case $S_1 = S_2$. Its scope is intentionally narrow. A larger network may change the solvency of bank 2 or create feedback through other defaulting nodes. Those regime changes are not represented by (18).

Remark 4.2 (Admissible balance-sheet changes). Changing entries of L without the corresponding transfer of assets creates nominal claims from nothing. Such an operation can alter proportional shares and can look profitable, but it is not a loan. Any strategic model should require double-entry balance-sheet consistency for every action.

5 Selective early repayment

5.1 Fixed-regime setup

Let bank 0 be insolvent and let its creditors be $1, \dots, m$. Write

$$P = \sum_{i=1}^m L_{0i} \quad (25)$$

for its initial liabilities. Before clearing, bank 0 makes early repayments x_i satisfying

$$0 \leq x_i \leq L_{0i}, \quad X := \sum_{i=1}^m x_i \leq e_0. \quad (26)$$

Let I denote the fixed incoming payment to bank 0 from the rest of the network. Assume throughout this section that the creditors remain solvent and that the other banks' payments do not change as x varies.

After repayment, bank 0 has

$$E = e_0 - X, \quad D = P - X, \quad (27)$$

where E is its remaining external asset and D its remaining nominal debt. If bank 0 was initially 0-insolvent, then

$$e_0 + I < P. \quad (28)$$

Subtracting X from both sides gives $E + I < D$. Early repayment cannot make bank 0 solvent by itself.

The total clearing payment made by bank 0 on its remaining debt is

$$W = \alpha E + \beta I. \quad (29)$$

Since $\alpha, \beta \leq 1$, one has $W < D$. Define the remaining liability share and effective recovery rate by

$$q_i = \frac{L_{0i} - x_i}{D}, \quad r = \frac{W}{D}. \quad (30)$$

Then $q_i \in [0, 1]$, $\sum_i q_i = 1$, and $r < 1$.

Up to terms independent of x , creditor i has value

$$R_i(x) = x_i + W q_i + K_i, \quad (31)$$

where K_i collects its external assets, other incoming payments, and liabilities.

5.2 Own and cross effects

Theorem 5.1 (Marginal effects of selective repayment). *Within the fixed default regime,*

$$\frac{\partial R_i}{\partial x_i} = 1 - \alpha q_i - r(1 - q_i) \geq 0, \quad (32)$$

$$\frac{\partial R_i}{\partial x_k} = q_i(r - \alpha), \quad k \neq i, \quad (33)$$

$$\frac{\partial}{\partial x_k} \sum_{i=1}^m R_i = 1 - \alpha \geq 0. \quad (34)$$

Proof. For repayment to creditor i , both $L_{0i} - x_i$ and D have derivative -1 , while W has derivative $-\alpha$. Differentiating (31) gives

$$\begin{aligned} \frac{\partial R_i}{\partial x_i} &= 1 - \alpha \frac{L_{0i} - x_i}{D} - W \frac{D - (L_{0i} - x_i)}{D^2} \\ &= 1 - \alpha q_i - r(1 - q_i). \end{aligned}$$

The quantity subtracted from 1 is a convex combination of α and r , both of which are at most 1. This proves (32).

For $k \neq i$, the numerator $L_{0i} - x_i$ is fixed, while W and D both decrease. Hence

$$\frac{\partial R_i}{\partial x_k} = -\alpha q_i + \frac{W}{D} q_i = q_i(r - \alpha),$$

which is (33). Finally,

$$\sum_{i=1}^m R_i = X + W + \sum_{i=1}^m K_i, \quad (35)$$

and differentiation gives (34). $\square$

Equation (32) proves the early-repayment conjecture in the 2020 thesis under its stated fixed-regime assumptions. The selected creditor cannot be harmed by receiving more money early.

Equation (33) gives the missing distributional result. All creditors not selected for repayment move in the same direction:

- if $r < \alpha$, every non-selected creditor with a positive remaining claim loses;
- if $r = \alpha$, non-selected creditors are locally indifferent;
- if $r > \alpha$, every creditor weakly benefits.

Corollary 5.2 (Marginal Pareto criterion). *A selective early repayment is a Pareto improvement for the creditors if $r \geq \alpha$. If $r < \alpha$ and a non-selected creditor has a positive remaining claim, the repayment is not a Pareto improvement even though total creditor value rises by $1 - \alpha$ per unit.*

The conflict is therefore not between efficiency and all repayment. It is between aggregate recovery and the distribution of that recovery.

The preceding calculation extends from a payment to one creditor to an arbitrary infinitesimal repayment direction. Let $h_i \geq 0$ be the rate of repayment to creditor i and put $H = \sum_i h_i$.

Theorem 5.3 (The local Pareto cone). *Within the fixed default regime, the directional derivative of creditor i 's value is*

$$\dot{R}_i = (1 - r)h_i + (r - \alpha)Hq_i. \quad (36)$$

Consequently:

- (i) if $r \geq \alpha$, every nonnegative repayment direction is locally Pareto improving for the creditors;
- (ii) if $r < \alpha$ and $H > 0$, a direction is locally Pareto improving if and only if

$$\frac{h_i}{H} \geq q_i \frac{\alpha - r}{1 - r} \quad \text{for every } i. \quad (37)$$

The cone in (37) is nonempty. When $\alpha < 1$, the share of the marginal repayment not forced by the inequalities is $(1 - \alpha)/(1 - r)$.

Proof. Along the direction h , one has

$$\dot{W} = -\alpha H, \quad \dot{D} = -H, \quad \dot{q}_i = \frac{Hq_i - h_i}{D}. \quad (38)$$

Differentiating $R_i = x_i + Wq_i + K_i$ gives

$$\dot{R}_i = h_i - \alpha Hq_i + r(Hq_i - h_i),$$

which is (36). If $r \geq \alpha$, both terms in (36) are nonnegative. If $r < \alpha$, then $r < 1$, and rearranging $\dot{R}_i \geq 0$ gives (37). The lower bounds on the repayment shares sum to

$$\frac{\alpha - r}{1 - r} \sum_i q_i = \frac{\alpha - r}{1 - r} \leq 1. \quad (39)$$

The remaining share equals $(1 - \alpha)/(1 - r)$. $\square$

Theorem 5.3 gives an exact local compensation rule. When $r < \alpha$, a targeted repayment may still be safe, but excluded creditors must receive the lower-bound shares in (37). Any remaining share can be directed to the creditor whose rescue creates the largest network benefit.

5.3 A pro-rata safe harbor

Selective repayment changes liability shares. A natural alternative is to allow early repayment only in the original proportions.

Theorem 5.4 (Pro-rata early repayment). *Let*

$$x_i(t) = tL_{0i}, \quad 0 \leq t < 1, \quad tP \leq e_0. \quad (40)$$

Within the fixed default regime,

$$\frac{dR_i}{dt} = (1 - \alpha)L_{0i} \geq 0 \quad \text{for every creditor } i. \quad (41)$$

Thus pro-rata early repayment weakly benefits every creditor and preserves their relative claims.

Proof. The remaining debt is $D(t) = (1 - t)P$, and

$$q_i(t) = \frac{(1 - t)L_{0i}}{(1 - t)P} = \frac{L_{0i}}{P} \quad (42)$$

is constant. Also

$$W(t) = \alpha(e_0 - tP) + \beta I. \quad (43)$$

Substitution in (31) gives

$$R_i(t) = tL_{0i} + \frac{L_{0i}}{P}W(t) + K_i. \quad (44)$$

Differentiating proves (41). $\square$

Proposition 5.5 (Characterization by share preservation). *Among nonzero infinitesimal repayment directions, the pro-rata direction $h_i = Hq_i$ is the unique direction that leaves every residual liability share q_i unchanged.*

Proof. The derivative calculated in the proof of Theorem 5.3 satisfies

$$\dot{q}_i = 0 \iff h_i = Hq_i. \quad (45)$$

Applying this equivalence to every i proves the claim. $\square$

Theorem 5.4 suggests a simple rule: a standstill need not ban all early repayment. It can permit payments that preserve the original creditor proportions. Such payments capture the value otherwise lost through the factor α without preferring one creditor over another.

5.4 Rescuing a creditor exposed to contagion

Suppose creditor i would become insolvent because bank 0 defaults. With all other early repayments fixed, Equation (32) shows that R_i is nondecreasing in x_i . Consequently the exact rescue test is an endpoint calculation:

$$R_i(x_i^{\max}) \geq 0, \quad x_i^{\max} = \min \left\{ L_{0i}, e_0 - \sum_{k \neq i} x_k \right\}. \quad (46)$$

If the inequality fails, no feasible selective repayment to i can rescue it within the fixed regime. If it holds, continuity gives a least rescue threshold. This formulation is more precise than a sufficient inequality that ignores the creditor's other assets and liabilities.

6 The continuous prepayment game

6.1 Strategies and balance-sheet updates

Fix an initial debt network (e, L) . Bank i chooses a vector of outgoing prepayments

$$x_i = (x_{ij})_{j \in \mathcal{N}} \quad (47)$$

from the compact polytope

$$\mathcal{X}_i = \left\{ x_i : 0 \leq x_{ij} \leq L_{ij}, \quad \sum_j x_{ij} \leq b_i, \quad 0 \leq b_i \leq e_i. \right\}, \quad (48)$$

Using an initial liquidity budget b_i makes the joint strategy space the product $\mathcal{X} = \prod_i \mathcal{X}_i$. In particular, one bank cannot fund its simultaneous strategy using a prepayment that it receives from another bank in the same action round.

For a profile $x \in \mathcal{X}$, double-entry consistency gives

$$e_i^x = e_i - \sum_j x_{ij} + \sum_j x_{ji}, \quad (49)$$

$$L_{ij}^x = L_{ij} - x_{ij}, \quad (50)$$

$$\bar{p}_i^x = \sum_j L_{ij}^x. \quad (51)$$

Thus $\sum_i e_i^x = \sum_i e_i$: prepayment transfers external assets but does not create them. Let Π^x be the relative-liability matrix constructed from L^x , and let $p^\uparrow(x)$ be the greatest Rogers–Veraart clearing vector of the updated network.

Two natural utility functions are total assets and equity:

$$A_i(x) = e_i^x + \sum_j \pi_{ji}^x p_j^\uparrow(x), \quad (52)$$

$$V_i(x) = [A_i(x) - \bar{p}_i^x]^+. \quad (53)$$

Total assets distinguish outcomes for an insolvent bank whose equity is zero. Equity is the direct objective of shareholders. Since these objectives can produce different games, every equilibrium statement should specify which one is being used.

6.2 Regular default regions

Let D be a proposed set of defaulting banks and let $S = \mathcal{N} \setminus D$. Consider an open set of prepayment profiles on which all banks in D are strictly insolvent, all banks in S are solvent, and $\bar{p}_i^x > 0$ for every relevant row. On this set,

$$p_D(x) = \alpha e_D^x + \beta (\Pi_{DD}^x)^\top p_D(x) + \beta (\Pi_{SD}^x)^\top \bar{p}_S^x, \quad p_S(x) = \bar{p}_S^x. \quad (54)$$

Define the default-set resolvent

$$M_D(x) = \left[I - \beta (\Pi_{DD}^x)^\top \right]^{-1} \quad (55)$$

whenever the inverse exists. It always exists when $\beta < 1$ because Π_{DD}^x is substochastic. When $\beta = 1$, invertibility is a regularity condition on the defaulting subnetwork.

Theorem 6.1 (Geometry of a fixed-default region). *On a region satisfying the preceding strict inequalities and invertibility condition,*

$$p_D(x) = M_D(x) \left[\alpha e_D^x + \beta (\Pi_{SD}^x)^\top \bar{p}_S^x \right]. \quad (56)$$

The clearing payments and the utilities in (52)–(53) are rational, hence real analytic, functions of x on the region. They need not be affine.

If every active debtor prepays proportionally,

$$x_{ij} = t_i L_{ij}, \quad 0 \leq t_i < 1, \quad (57)$$

then $\Pi^x = \Pi$. On every fixed-default region, $p^\uparrow(x)$ is then an affine function of the vector t .

Proof. Equation (56) follows by rearranging (54). Assets and nominal liabilities are affine in x . Every nonzero relative-liability entry has the form

$$\pi_{ij}^x = \frac{L_{ij} - x_{ij}}{\bar{p}_i - \sum_k x_{ik}}, \quad (58)$$

so it is rational. Matrix inversion expresses each entry of $M_D(x)$ as a ratio of determinants. Formula (56) is therefore rational wherever the denominators and determinant do not vanish. Selective prepayment generally changes the displayed ratio nonlinearly, which rules out a general affine claim.

Under (57), every residual row of liabilities is multiplied by $1 - t_i$, so its relative proportions are unchanged. The matrix in (55) is constant, while e^x and $\bar{p}^x$ are affine in t . Formula (56) is therefore affine. $\square$

Theorem 6.1 gives a finite default-set decomposition, but it is not a globally smooth decomposition. Default costs can create jumps when a bank becomes just solvent. Such boundaries can determine best responses, as Theorem 6.3 below shows.

6.3 A resolvent formula for marginal effects

Let $x(t) = x + th$ remain inside a regular fixed-default region for small t . Put

$$H_i = \sum_j h_{ij}. \quad (59)$$

Dots denote derivatives at $t = 0$. Direct differentiation of the balance-sheet update gives

$$\dot{e}_i = -H_i + \sum_j h_{ji}, \quad (60)$$

$$\dot{\bar{p}}_i = -H_i, \quad (61)$$

$$\dot{\pi}_{ij} = \frac{\pi_{ij}^x H_i - h_{ij}}{\bar{p}_i^x}. \quad (62)$$

Theorem 6.2 (Prepayment Jacobian). *Inside a regular region with default set D , the directional derivative of the clearing vector is $\dot{p}_S = \dot{\bar{p}}_S$ and*

$$\dot{p}_D = M_D(x) \left[\alpha \dot{e}_D + \beta (\dot{\Pi}_{DD})^\top p_D + \beta (\dot{\Pi}_{SD})^\top \bar{p}_S + \beta (\Pi_{SD}^x)^\top \dot{\bar{p}}_S \right]. \quad (63)$$

Consequently,

$$\dot{A}_i = \dot{e}_i + \sum_j \left(\dot{\pi}_{ji} p_j + \pi_{ji}^x \dot{p}_j \right). \quad (64)$$

Proof. Differentiate (54) and collect the terms containing $\dot{p}_D$ on the left. Multiplication by the inverse in (55) gives (63). Equation (64) follows by differentiating (52). $\square$

The resolvent M_D is a network multiplier. The bracketed vector in (63) is the direct balance-sheet and liability-share effect of the action; multiplication by M_D propagates it through cycles of defaulting banks. A local best-response calculation therefore requires one factorization of $I - \beta(\Pi_{DD}^x)^\top$ and one linear solve per candidate direction.

6.4 A uniquely optimal fractional prepayment

The discrete prepayment model permits only full repayment of an edge or no repayment. The following example shows that this restriction can exclude the unique best response.

Theorem 6.3 (Fractional best responses are essential). *For every $\alpha \in [0, 1)$, there is a Rogers–Veraart network with $\beta = 0$ in which a bank maximizing total assets has a unique best response that prepays a strict fraction of one liability. The same conclusion holds for every $\beta \in [0, 1)$ when $\alpha = 0$.*

Proof. Consider banks 0, 1, 2 and an outside sink ω . Let

$$L_{01} = 10, \quad L_{0\omega} = 30, \quad L_{12} = 10, \quad L_{20} = 10, \quad (65)$$

with all other liabilities zero. External assets are

$$e_0 = 10, \quad e_1 = 4, \quad e_2 = e_\omega = 0, \quad (66)$$

and first set $\beta = 0$. Only bank 0 acts. It chooses a prepayment $x = x_{01} \in [0, 10]$. Bank 0 is always insolvent and pays $p_0 = \alpha(10 - x)$. Its clearing payment to bank 1 is

$$z_{01}(x) = \alpha \frac{(10 - x)^2}{40 - x}. \quad (67)$$

Bank 1 becomes solvent at the unique solution x_α of

$$4 + x + \alpha \frac{(10 - x)^2}{40 - x} = 10. \quad (68)$$

Indeed, the left-hand side is strictly increasing on $[0, 10]$. To see this, put $g(x) = (10 - x)^2/(40 - x)$. Then

$$g'(x) = \frac{(10 - x)(x - 70)}{(40 - x)^2} \geq -\frac{7}{16}, \quad (69)$$

so the derivative of the left-hand side of (68) is at least $1 - 7\alpha/16 > 0$. Its value at $x = 0$ is $4 + 5\alpha/2 < 10$, while its value at $x = 6$ is $10 + 8\alpha/17 \geq 10$. Hence

$$0 < x_\alpha \leq 6 < 10. \quad (70)$$

Below x_α , bank 1 defaults and bank 2 receives less than its nominal claim. Since $\beta = 0$ and bank 2 has no external assets, it pays zero to bank 0. At and above x_α , bank 1 pays 10 to bank 2, which becomes solvent and pays 10 to bank 0. Bank 0 remains insolvent because its post-action liabilities are $40 - x$ and its pre-haircut resources are at most $20 - x$. Bank 0's total-asset utility is therefore

$$A_0(x) = \begin{cases} 10 - x, & 0 \leq x < x_\alpha, \\ 20 - x, & x_\alpha \leq x \leq 10. \end{cases} \quad (71)$$

Its unique maximizer is $x = x_\alpha$, a strict fraction of L_{01} .

Finally, set $\alpha = 0$ and take any $\beta < 1$. For $x < 6$, the payments in the cycle satisfy

$$p_0 = \beta p_2, \quad p_1 = \beta \frac{10 - x}{40 - x} p_0, \quad p_2 = \beta p_1. \quad (72)$$

Their only solution is zero because $\beta^3(10 - x)/(40 - x) < 1$. At $x = 6$, bank 1 and then bank 2 become solvent. The same utility calculation gives a unique maximizer $x = 6$. $\square$

The optimum in this example is a solvency threshold. Below it, the marginal payment does not activate the return path $1 \rightarrow 2 \rightarrow 0$; above it, further prepayment only spends bank 0's external assets. This phenomenon is missed both by endpoint-only strategies and by optimization confined to the interior of a fixed-default region.

7 A precise standstill problem

The word “fraud” in the working notes covers two different phenomena. Creating a liability without transferring the corresponding asset violates balance-sheet consistency. Selective early repayment is different: it is a real transfer, but it violates equal treatment of creditors.

The latter creates a simple coalition incentive.

Proposition 7.1 (Bilateral repayment incentive). *Suppose bank 0 remains insolvent, so its post-clearing equity is zero. If bank 0 and creditor i can jointly choose x_i at no transaction cost while other repayments are fixed, then the coalition weakly prefers the largest feasible x_i . The preference is strict whenever creditor i 's derivative in (32) is positive.*

Proof. Bank 0 remains 0-insolvent because early repayment subtracts the same amount from its assets and liabilities. Its equity therefore remains zero. The selected creditor's value is nondecreasing by (32). $\square$

This proposition captures the valid part of the tentative standstill theorem. It is not by itself a Nash theorem. A Nash statement requires a specified action protocol, control rights over payments, liquidity constraints, information, side payments, and the actions available to excluded creditors.

The results suggest three possible institutional rules.

- (a) **Strict standstill.** Freeze all pre-clearing transfers. This prevents preference but discards the surplus from avoiding liquidation.
- (b) **Pro-rata safe harbor.** Permit only repayments proportional to existing liabilities. Theorem 5.4 shows that this is a Pareto improvement for creditors in the fixed regime.
- (c) **Pareto-cone safe harbor.** Permit a targeted repayment when its marginal shares satisfy (37). The uncommitted share $(1 - \alpha)/(1 - r)$ can target a creditor near insolvency while the required lower-bound shares protect the others. Implementation requires observable claims and an estimate of r .

8 Nonlinear liquidation losses

A constant recovery fraction is a useful first model, but market impact should usually depend on the amount liquidated. Let $F(E)$ be the cash recovered from liquidating external assets E , with

$$0 \leq F(E) \leq E. \quad (73)$$

The linear model is $F(E) = \alpha E$. Under pro-rata early repayment,

$$W(t) = F(e_0 - tP) + \beta I. \quad (74)$$

Proposition 8.1 (Pro-rata repayment with market impact). *If F is differentiable, then*

$$\frac{dR_i}{dt} = L_{0i}[1 - F'(e_0 - tP)]. \quad (75)$$

Pro-rata early repayment benefits every creditor whenever the marginal liquidation recovery satisfies $F'(E) \leq 1$.

Proof. The liability share remains L_{0i}/P . Therefore

$$R_i(t) = tL_{0i} + \frac{L_{0i}}{P}[F(e_0 - tP) + \beta I] + K_i. \quad (76)$$

Differentiate with respect to t . □

The economically relevant primitive is thus the total recovery function F , not merely a recovery fraction written as $\alpha(E)$. If $F(E) = \alpha(E)E$, then

$$F'(E) = \alpha(E) + E\alpha'(E). \quad (77)$$

9 Discrete time

For a dynamic version, the repayment budget must be aggregate. At date t , choose $x_i(t) \geq 0$ such that

$$\sum_i x_i(t) \leq e_0(t), \quad x_i(t) \leq L_{0i}(t). \quad (78)$$

The balance-sheet updates are

$$e_0(t+1) = e_0(t) - \sum_i x_i(t), \quad (79)$$

$$e_i(t+1) = e_i(t) + x_i(t), \quad (80)$$

$$L_{0i}(t+1) = L_{0i}(t) - x_i(t). \quad (81)$$

These equations conserve external assets inside the network. If incoming payments to bank 0 are unchanged, its pre-haircut solvency gap is invariant:

$$[P(t+1) - e_0(t+1) - I(t+1)] = P(t) - e_0(t) - I(t). \quad (82)$$

Thus repayment alone cannot repair bank 0's own fundamental deficit, though it can alter the solvency of its creditors and hence future feedback through the network.

A complete dynamic game would also specify the clearing time, stochastic external-asset shocks, the timing of information, and whether actions are observable. Without these ingredients, claims about equilibrium timing are not well defined.

10 Research programme

The calculations above lead to several concrete problems.

10.1 A global best-response algorithm

Theorem 6.2 solves the smooth part of the problem. A global algorithm should combine three types of candidates:

- (i) stationary points and active faces of the strategy polytope inside a regular default region;
- (ii) solvency boundaries at which a bank changes branch in the clearing map;
- (iii) singular boundaries at which a liability row vanishes or the default-set resolvent ceases to exist.

Theorem 6.3 shows that the second class cannot be omitted. An implementable first algorithm could enumerate default sets for small networks, use (63) for local optimization, and solve one-dimensional threshold equations along candidate edges.

10.2 Existence of equilibrium

The strategy space (48) is compact and convex, but default costs can make the utilities discontinuous at solvency boundaries. Compactness alone therefore does not give a pure or mixed equilibrium. Useful next results would identify network classes for which best responses are nonempty and convex-valued, or replace the discontinuous default rule by a continuous recovery function and apply a fixed-point theorem. A separate question is whether equilibrium is restored when actions must preserve liability proportions.

10.3 Robust fractional optimality

Theorem 6.3 covers both recovery-parameter axes. The next step is to characterize fractional threshold optima when $\alpha, \beta > 0$. One expects them when the return through a newly solvent path exceeds the direct cost of reaching the threshold, while the marginal return beyond the threshold is below one. A useful theorem would state these conditions in terms of the two neighboring default-set resolvents.

10.4 Strategic games with explicit control rights

To turn the standstill intuition into an equilibrium theorem, specify:

- (i) who proposes and authorizes a repayment;
- (ii) whether creditor consent is required;
- (iii) whether side payments are allowed;
- (iv) what each player observes;
- (v) whether new liabilities require a matching asset transfer;
- (vi) how the clearing date is chosen.

The bilateral incentive proposition then becomes an input to the game rather than a substitute for its definition.

10.5 Mechanism design for safe early repayment

The local Pareto cone is the simplest targeted rule. Further work should ask whether its stepwise application remains Pareto improving after the effective recovery rate and the default set change. If selective repayment prevents a specific contagion event, the mechanism should compare the jump in network value at that boundary with the compensation required along the path to it. The pro-rata rule remains the information-light benchmark because it does not require an estimate of r .

10.6 Dynamic and coalitional extensions

With multiple action dates, a bank may fund a later prepayment from an earlier receipt. The game then needs explicit settlement order, information, and clearing time. Coalitions introduce a second problem: the debtor may have zero equity and be indifferent while a selected creditor gains. Side payments can turn that weak incentive into a strict joint deviation. A promising solution concept is therefore coalition-proof equilibrium under the Pareto-cone safe harbor rather than Nash equilibrium under unrestricted actions.

11 Conclusion

The notes point to a genuine design tension. A complete payment freeze protects equal treatment but can force avoidable liquidation losses. Unrestricted pre-clearing bargaining can preserve value but allows coalitions to redirect recovery away from excluded creditors.

The continuous model sharpens this tension. Inside a regular default region, the resolvent formula makes marginal incentives computable. Selective actions produce rational rather than generally affine payoffs, while proportional actions recover an affine structure. At region boundaries, fractional payments can be uniquely optimal because the least payment that rescues a bank activates an entire return path through the network.

These results support a middle course for a clearing protocol: enforce balance-sheet consistency, use pro-rata repayment when only local information is available, and permit targeted departures when the Pareto-cone constraints can be verified. The main mathematical problem left is global: combine smooth optimization inside default regions with the discrete solvency transitions between them, then establish equilibrium under the resulting discontinuous payoffs.

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